cansats in europe

CanSat 2026/2027
Preliminary Design Review (PDR)
Team: SkipSat
Country: Poland

Table of Contents

1 INTRODUCTION 3
- 1.1 Team organisation and roles 3
- 1.2 Mission objectives 5

2 CANSAT DESCRIPTION 7
- 2.1 Mission overview 7
- 2.2 Mechanical/structural design 8
- 2.3 Electrical design 9
- 2.4 Software design 12
- 2.5 Recovery system 15
- 2.6 Ground support Equipment 15

3 PROJECT PLANNING 16
- 3.1 Time schedule 16
- 3.2 Task list 16
- 3.3 Resource estimation 16

4 OUTREACH PROGRAMME 17

1 Introduction

1.1 Team organisation and roles

Teacher: Mariola Jachowicz

Team members:

1) 🇵🇱 Mikołaj Szadkowski:
   a) Finalist of the National Olympiad in Informatics.
   b) Passionate about physics, algorithmics, and C++ programming.
   c) Contributor to SciCalc.
   d) Responsible for:
      i) Physics calculations and simulations, including dynamic and control modeling.
      ii) Algorithm development and analysis for flight control systems.
      iii) Programming in C++.

2) 🇵🇱 Jan Pawlak: **(Team leader)**
   a) Finalist of National Junior Informatics Olympiad. Passionate about electronics and industrial automation.
   b) 5 years of experience in 3d CAD
   c) Responsible for:
      i) systems programming including flight software, sensor data processing, and telemetry communication.
      ii) Building and testing prototypes
      iii) Electronics and hardware design, mechanical design

3) 🇵🇱 Kuba Sterna:
   a) Under 4 hour marathon runner.
   b) Passionate about statistics and movies.
   c) Responsible for:
      i) Social media marketing
      ii) Sponsorship arrangements
      iii) Popularizing the knowledge of the project

4) 🇵🇱 Kacper Chabowski :
   a) Passionate about mathematics and control theory.
   b) Responsible for:
      i) Systems programming including flight software, sensor data processing, and telemetry communication.
      ii) Ground station software development
      iii) Research and development of flight control systems .

5) 🇵🇱 Kajetan Sosnowski :
   a) Laureate of National Junior Maths Olympiad
   b) Passionate about astrophysics, C++ programming;
   c) Responsible for:
      i) Social media marketing
      ii) Sponsorship arrangements
      iii) Popularizing the knowledge of the project
      iv) Building and testing prototypes
      v) Helping other team members with their primary responsibilities

1.2 Mission objectives

The primary mission of our CanSat is to measure atmospheric pressure and temperature throughout its descent, transmit the data to the ground station in real time, and store it locally on the CanSat's onboard memory. Using these measurements, we can calculate the altitude and construct a detailed vertical profile of the atmosphere.

Our secondary mission is deploying navigation and communication infrastructure without satellites. The CanSat separates mid-air into one MainSat and three MiniSats (four if mass and volume allow); together we call them the units. Each unit steers its parachute towards its own landing coordinates by pulling the lines connected to the parachute. During the flight, the MiniSats will establish and maintain stable communication with the MainSat. We also plan on using trilateration (determining a position from measured distances to reference points) to determine the position of the MainSat relative to the MiniSats throughout the entire mission. Since the units steer towards targets spread apart from each other, the trilateration will be more accurate. After landing, the units will form a mesh network in which every unit relays the

messages of the others, and the MiniSats will serve as trilateration reference points. We will also try to operate the mesh network already during the flight.

The main engineering problem is fitting an active, steered landing system into units this small. Trilateration and the mesh network are both enabled by the choice of the radio module: one module that supports both distance measurement and normal data communication provides both functions.

This system could be used on planets without satellite navigation systems to locate objects relative to the MiniSats, and to deploy basic communication infrastructure.

Each MiniSat can complete the primary mission on its own, which provides additional redundancy.

Mission Success Criteria:

- Primary Mission:
  - At least one unit (the MainSat or a MiniSat) measures atmospheric pressure and temperature continuously during descent.
  - The collected data is successfully transmitted to the ground station in real time and stored locally on the onboard memory.
  - The obtained results are consistent with expected trends from the International Standard Atmosphere within acceptable error margins.
- Secondary Mission
  - The GPS, accelerometer, and gyroscope of every unit record valid data throughout descent and landing.
  - The MainSat performs a controlled descent and lands within a sufficient radius of its target coordinates for the planned trilateration.
  - All telemetry and onboard data are successfully received or recovered for post-flight analysis.
  - The MiniSats separate mid-air, and land without significant damage.
  - The MiniSats perform a controlled descent, during which they accumulate atmospheric pressure and temperature data, followed by a landing within a sufficient radius of their target coordinates for the planned trilateration.
  - After landing, the MiniSats and the MainSat form a mesh network in which every unit relays the messages of the others.
  - Optional: the mesh network also operates during the flight.

- All units are able to locate themselves relative to one another by trilateration.

2 CANSAT DESCRIPTION

2.1 Mission overview

Our CanSat will be launched by a rocket to an altitude between 1.5 km and 2.5 km, where it will be ejected and begin its descent. The descent rate will remain within the required range of 5 - 12 m/s. During this phase, the CanSat will carry out the primary mission, continuously recording and transmitting atmospheric pressure and temperature data to the ground station while storing all measurements locally for redundancy.

From these measurements, the altitude will be derived, allowing us to construct a vertical profile of the atmosphere. By comparing the gathered data with calculated values, we will evaluate how local conditions deviate from theoretical models and how temperature and pressure vary with height. This information will help us better understand the structure of the lower atmosphere at the time of launch.

A set time after ejection, if the altitude is still above a minimum safe value and the conditions are safe, a release mechanism separates the MiniSats from the MainSat. From then on, each unit steers independently towards its own landing coordinates, so that the units land in positions suitable for trilateration.

Each unit is designed to land as close as possible to its own target coordinates, which are sent by radio from the ground station before launch and can be updated during the flight, depending on how the rocket flies; different units can have different targets. The targets will be placed on slight hills: after landing the antennas are very close to the ground, and an elevated landing spot reduces signal obstruction between the units and towards the ground station. Steering will be performed by pulling the parachute control lines, allowing mid-air directional adjustments. This approach is inspired by this research paper on steerable parachute systems.

To support this, each unit includes a GPS module, MEMS accelerometer, and gyroscope to measure position, acceleration, and orientation. These sensors will provide data for analyzing the descent trajectory and stability. By combining GPS and IMU data, we will reconstruct the full flight path, assessing how accurately each unit can navigate.

During the flight, the units communicate with each other and use trilateration to determine their positions relative to each other.

After landing, the units will operate as a mesh network: data from every unit can reach the ground station through the other units, even if that unit has no direct link to it. If there is enough space, the units will also be equipped with solar panels, so that they could run for longer periods of time for instance if they were deployed on another planet.

Overall, the primary mission focuses on accurate atmospheric data collection and model validation, while the secondary mission demonstrates deploying navigation and communication infrastructure without satellites: steered landing of the units, relative positioning by trilateration and a mesh communication network. Together, these missions showcase both scientific measurement capability and advanced robotic functionality within the CanSat framework.

We chose this secondary mission because it presents valuable engineering challenges similar to those encountered in real space exploration. It could be used to locate objects relative to the MiniSats. This type of technology, which allows objects to determine their positions without relying on satellite communication, could be useful during the early stages of settling on new planets. Dropping the MiniSats is also an example of deploying communication infrastructure on another planet: their mesh network could be used by other objects as reference points to determine their own positions and to relay their data. Beyond these applications, the topic is of strong interest to our team, and we want to find out how much of it can be achieved within the constraints of a CanSat.

2.2 **Mechanical/structural design**

The MainSat will be a mostly 3D printed frame carrying a custom PCB, which also provides the pogo pin connection to the top MiniSat. The MainSat also holds a larger battery that powers the MiniSats until deployment, so that the MiniSat batteries can be smaller (see 2.3). The top and bottom of every MiniSat will be FR-4 PCBs. They carry the pogo pin connections of the stack and, because they are also structural parts, they give us board space for the sensors, modules and MCU without adding extra structure. The rest of each MiniSat will be 3D printed and, if required, CNC machined from aluminium.

All 3D printed parts will be manufactured using FDM technology from PLA filament. PLA offers suitable mechanical strength at temperatures below 50 °C, which will not be exceeded during operation. To secure additional components and PCB assemblies,

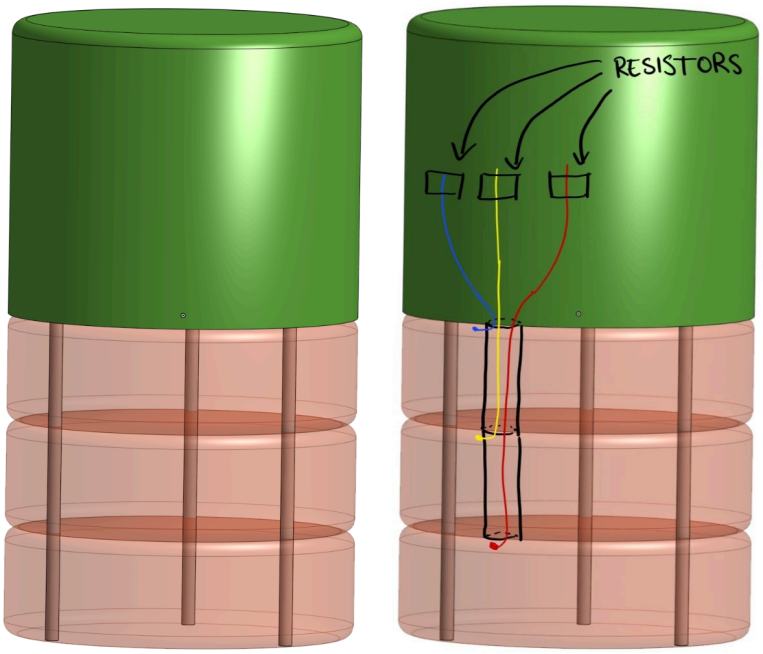
small metric screws will be used together with heat-set inserts embedded directly into the printed frame, ensuring structural stability under high g-forces.

The parachute control system follows the steering concept of the paper linked in 2.1, but the control lines are pulled by four thin nitinol springs instead of motors. Nitinol is a shape memory alloy: when heated, the spring contracts and pulls its control line. Each spring is heated by running a current through it, controlled by the MCU (see 2.3). When the current is switched off, the spring is cooled by the airflow and the tension of the parachute line stretches it back; an additional line prevents the spring from being overstretched. Most lines, including the parachute control lines, will be fishing lines.

To comply with weight restrictions, lightweight materials such as thermoplastics, FR-4 PCBs, and aluminum alloys will be employed. The total mass of the CanSat, including all units, antennas and parachutes, will be kept within the required 300–350 g (section 6.2.1d of the competition regulations). However, if the total mass remains significantly below the allowable limit, strategically positioned ballast weights will be added to enhance flight maneuverability.

Before separation, the MiniSats are stacked below the MainSat, each with its parachute packed on its top (see the picture below). One to three tunnels run through the stack from the bottom MiniSat up to the MainSat. Each MiniSat has its own tunnel sections, which are not joined to those of its neighbours, so the stack can come apart. The MiniSats are held by lines that run through the tunnels up to the MainSat. The spring-loaded pogo pins between neighbouring MiniSats push them apart and keep these lines in tension; the same pogo pins carry power and data (see 2.3). Each

MiniSat is released separately: its lines are cut by heating them with its own resistor. The released MiniSat falls away and its parachute unfolds automatically.



MainSat (green) with three MiniSats stacked below it; the rods are the line tunnels

2.3 Electrical design

- **Architecture.**
  We will design custom PCBs for the MainSat and the MiniSats. The mission requires more functionality than the CanSat Kit provides, and the compact MiniSats need boards whose mounting holes and other features are closely integrated with the frame. The microcontroller will most likely be from the STM32 family.
  For onboard data storage, an SPI Flash chip soldered directly to the PCB will be used due to its superior vibration resistance and overall reliability compared to SD cards. However, since Flash chips are typically limited to capacities below 32 MB, a microSD card will be employed to store data received from the other units, along with sparse data, and a copy of the flash chip, for redundancy.
  The MiniSats are connected by pogo pins until deployment; through them,

they receive power and communicate with the rest of the system. Loss of the pogo pin contact also confirms that a MiniSat has been deployed.

- **Primary mission sensors.**
  Every unit will measure temperature with an LM35 thermometer and pressure with a BMP280 barometer, the same sensors as in the CanSat Kit. The kit contains only one of each, so the sensors for the other units will be bought separately. This way every MiniSat can perform the primary mission on its own. The LM35 needs a supply voltage of at least 4 V, which a single Li-Ion cell does not always provide, so we will either power it from a step-up converter or replace it with a temperature sensor that works at 3.3 V. Additional temperature readings are available from the sensors integrated into other modules, e.g. the barometer and the accelerometer.

- **Secondary mission devices.**
  The MainSat and every MiniSat will have their own radio, GPS and a parachute control system, consisting of four nitinol springs. The heating current of each spring is switched by the MCU through a MOSFET. The separation system requires only the release lines and the resistor heating circuit described in 2.2.

- **Power supply.**
  Every unit will be powered by a single-cell lithium battery. We prefer Li-Po batteries, because they are often sold in form factors that fit the small units better than Li-Ion cells. Using Li-Po batteries requires the organizers' approval (section 6.4.4 of the competition regulations), so we will send a request; if it is not approved, we will find other batteries, e.g. Li-Ion, that fit. The MainSat will have a larger battery that powers the MiniSats through the pogo pins until deployment, so the MiniSat batteries only need to cover the time after separation and can be smaller. The battery capacities will be chosen from a power budget, which we will prepare once the components are selected.

- **Communication system.**
  Every unit will be equipped with its own transceiver allowing the units to

communicate with each other. The link with the ground station is bidirectional: all units send telemetry to the ground station, and the ground station sends each unit its target landing coordinates before launch and can update them during the flight, depending on how the rocket flies. After landing, and if possible already during the flight, the units form a mesh network: every unit relays the messages of the others, so data from a unit without a direct link to the ground station still reaches it.

The simplest approach to trilateration uses the RSSI (Received Signal Strength Indication) of messages received from the other units and converts it to a distance with a signal strength model. Under ideal conditions, with links to all three other units, this gives approximate distances. Combined with the knowledge that the MainSat is above the MiniSats, this is enough to determine its position. A more advanced approach uses radios with TOF (Time Of Flight) ranging, which measures distance much more accurately and makes the trilateration correspondingly more precise. We prefer the second approach: the Semtech LR2021 transceiver meets our other requirements and supports both round-trip time-of-flight ranging and normal LoRa communication, so the same radio provides the distances for trilateration and the links of the mesh network. It covers 150–960 MHz, so it can operate in the 433 MHz ISM channel assigned by the organizers.

The LR2021 is not one of the radio modules allowed by default (SX127x, RFM9x or the CanSat Kit radio), so we will use it only if the organizers approve it, as required by section 6.2.3c of the competition regulations. We will send this request and expect it to be accepted, because the LR2021 uses the same 433 MHz band and LoRa modulation compatible with the default modules. If it is not approved, we will use a default module and base the trilateration on the RSSI approach described above.

2.4 **Software design**

Overview:

- The flight software of each unit manages sensor data, actuators and radio communication. It supports the primary mission (atmospheric pressure and temperature profiling) and the secondary mission (separation, guided landing and trilateration between the units). It will be written in C++, the language our team knows best, which is well suited to embedded programming because it gives precise control over memory and execution.
- GitHub link: https://github.com/richardead/SkipSat

Software Modes:

- Pre-Launch Mode: Remain armed, receive mission time, target landing coordinates and separation parameters by radio, calibrate altitude to zero, log at a low rate to save battery.
- Ascent Mode: Detect launch and peak altitude from the sensors, log continuously, begin telemetry transmissions.
- Descent Mode: Primary mission - sample pressure and temperature, derive altitude, log data on board and transmit the current values. Parachute not steered. Accept target coordinate updates from the ground station.
- Separation Mode: Release each MiniSat separately at the set time after ejection, if the altitude is above the minimum safe value, and confirm from current, contact and motion that each MiniSat has left.
- Guided Descent Mode: Navigate using GPS+IMU, estimate wind and descent rate, apply target coordinate updates from the ground station, decide whether the target is reachable and steer the parachute with the nitinol springs. If the in-flight mesh network is enabled, relay the messages of the other units.
- Mesh Mode: After landing - keep logging and beaconing at a reduced rate, relay data between the units, locate them relative to each other by trilateration, manage the solar power duty cycle (if solar panels are fitted).
- Safe Mode: Entered on any fault - steering switched off, stable unsteered descent.

2.5 Recovery system

The MainSat and every MiniSat have their own parachute, steered by four nitinol springs as described in 2.2. Before separation, the whole stack descends on the MainSat's parachute. After separation, the MainSat descends more slowly than the MiniSats, and the descent rate of every unit stays within the required range of 5–12 m/s.

The target landing coordinates will be placed on slight hills (see 2.1) that are easily accessible and free of obstacles, to simplify recovery operations and make the trilateration more accurate. Throughout the flight, GPS data will be continuously transmitted to the ground station and stored onboard, allowing real-time tracking of the positions of all units. After landing, the final GPS coordinates will make it easy to accurately locate and retrieve all units, even if they drift away from their targets.

2.6 Ground support Equipment

Hardware

- Manual directional antenna on tripod: primary ground station link for telemetry and target coordinate updates – manually aimed.
- Experimental directional antenna on motorised tripod that can auto-point using live GPS data transmitted by the units – used in parallel for testing and redundancy.
- Two microcontroller + radio transceivers: one per antenna, forwarding telemetry to the laptop and sending target coordinates from the laptop to the units.
- Laptop with GUI and storage: real-time display (telemetry, map) and logging data from all the units.

3 Project Planning

3.1 Time schedule

- Mid October - Mid November: Concept / Detailed Design
- Mid November - Late December: Prototyping
- Early January - Mid February: Integration & Software Development
- Mid February - 21 March: Tests, Design Iterations, Final Design
- Late March - Mid April: Campaign Preparation
- 17–18 April 2027: Campaign / Launch

3.2 Task list

Task List SkipSat

3.3 Resource estimation

Our current estimate of the component costs is well below the 2500 PLN limit (section 6.3 of the competition regulations). A detailed budget will be presented in the CDR.

4 Outreach programme

We plan to start a YouTube channel where we will post updated videos every month summarizing our project's progress. Additionally we will post about major problems we encounter and their solutions taking the viewer along with us on the journey of problem solving and thinking like an engineer. The videos will cover the physical aspects of building and testing the CanSat, systems programming, and the theoretical planning of the CanSat's construction and mission. To keep viewers engaged, our videos will include simple animations and calm background music. We will also design a simple logo for our channel.

Continuing with social media, we will also make an X account, where we will be posting progress updates. Some progress could be without any necessary visual components, but would be interesting to our potential followers. This could also work as a mission in-progress logbook, as the posts would be parts of our mission progress. We also plan to start posting on other social media, such as YouTube Shorts, TikTok and Instagram Reels, by including the most interesting engineering problems and design solutions from the long form content to funnel people into the long form content to learn more about the program and the problem solving skills required to become an engineer.

We are also planning to give several presentations during physics classes. The goal is to introduce our classmates to the CanSat project and explain the plan for our specific mission. To stay aligned with the curriculum, we will focus on the physical aspects of our mission, such as the relationship between altitude and pressure. We will also explain how we plan to use the GPS, accelerometer, and gyroscope to ensure that the MainSat and MiniSats land at their target coordinates.